Why Do We Care About Nash Equilibrium?

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You Are a State: What Would You Do?

Two rival states choose **simultaneously** whether to increase military spending **without observing the other state's choice in advance**. (Higher numbers are better)

		State B	
		Don't Arm	**Arm**
State A:	**Don't Arm**	(3,3)	(1,4)
	Arm	(4,1)	(2,2)

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Scan to vote

Your task – 30 seconds

1. **Vote privately:** You are State A, Arm or Don't Arm?
2. **Turn to a partner:** Explain why you chose it.

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Learning goal: By the end, you should be able to identify a Nash equilibrium and explain *why* mutual best responses make it an equilibrium.

Best Responses: One Player at a Time

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,4)
A: Arm	(4,1)	(2,2)

First: State A

If State B chooses **not to arm**:

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If State B chooses **not to arm**:

$4 > 3 \Rightarrow$ **A should arm**

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A player's **best response** is their best strategy *given what the other player does*.

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A player's **best response** is their best strategy *given what the other player does*.

Your turn – 20 seconds

Now look at **State B's payoffs**.

What is B's best response if A does not arm? What if A arms?

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Now look at **State B's payoffs**.

What is B's best response if A does not arm? What if A arms?

In both cases, B's best response is also Arm.

From Best Responses to Nash Equilibrium

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,**4**)
A: Arm	(**4**,1)	(**2**, **2**)

Red = A's best response **Blue = B's best response**

Question

Where are **both players** best responding to each other?

From Best Responses to Nash Equilibrium

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,**4**)
A: Arm	(**4**,1)	(**2**, **2**)

Red = A's best response **Blue = B's best response**

Mutual best responses

$$\boxed{(\text{Arm}, \text{Arm})}$$

Both states are playing a best response to the other's choice.

From Best Responses to Nash Equilibrium

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,**4**)
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Nash equilibrium

A **Nash equilibrium** is a strategy profile in which **each player's strategy is a best response to the other player's strategy**.

(Arm, Arm) is a Nash equilibrium.

From Best Responses to Nash Equilibrium

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,**4**)
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20-second partner check

Why call this an **equilibrium**?

Starting from (Arm, Arm), can either state get a higher payoff by changing its choice **alone**?

From Best Responses to Nash Equilibrium

	B: Don't Arm	**B: Arm**
A: Don't Arm	(3,3)	(1,**4**)
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Why is it an equilibrium?

At (Arm, Arm):

- If A changes alone to **Don't Arm**: $2 \to 1$
- If B changes alone to **Don't Arm**: $2 \to 1$

No profitable unilateral deviation

That is the sense in which the outcome is strategically stable.

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Scan to vote

Step 1: Vote individually

How many Nash equilibria does this game have?

A. 0 **B. 1** **C. 2** **D. 4**

Use the idea of **mutual best responses**.

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Step 2: 20-second partner discussion

Compare your answers.

If you disagree, try to convince your partner by asking:

What is Player 1's best response?
What is Player 2's best response?

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Step 3: Vote again

After your discussion, answer again:

A. 0 **B. 1** **C. 2** **D. 4**

Did your answer change?

Vote again

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Answer

There are **two Nash equilibria**:

$\boxed{(\text{Up}, \text{Left})}$ $\boxed{(\text{Down}, \text{Right})}$

At each outcome, **both players are best responding to each other**.

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Exit check

Complete the sentence with your partner:

"A Nash equilibrium is stable because . . ."

Your Turn: Find the Equilibrium

	Player 2: Left	**Player 2: Right**
Player 1: Up	(3,3)	(0,0)
Player 1: Down	(0,0)	(1,1)

Takeaway

A Nash equilibrium is stable because

no player can improve by changing strategy alone

In other words:

mutual best responses $\iff$ no profitable unilateral deviation